A proof of the conjectured linear recurrence and generating function for OEIS A298282

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1 The conjecture

The Online Encyclopedia of Integer Sequences records A298282 as

> *Number of n X 3 0..1 arrays with every element equal to 0, 1, 2, 3 or 6 king-move adjacent elements, with upper left element zero*

and carries in its Formula section the lines:

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Empirical:  a(n) = 3*a(n-1) - a(n-2) + 2*a(n-3) + 4*a(n-4) - a(n-5) + 2*a(n-6)
- 2*a(n-7) - 3*a(n-8) - 2*a(n-9) - 2*a(n-10) - 2*a(n-11) for n>12.

Empirical g.f.:  x*(4 + 13*x - x^2 + 12*x^3 + 3*x^4 - 13*x^5 - 8*x^6 -
19*x^7 - 13*x^8 - 7*x^9 - 2*x^10 - 4*x^11) / (1 - 3*x + x^2 - 2*x^3 - 4*x^4
+ x^5 - 2*x^6 + 2*x^7 + 3*x^8 + 2*x^9 + 2*x^10 + 2*x^11).  - Colin Barker,
Feb 26 2018
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That line carries no separate signature; the entry itself is by R. H. Hardin (Jan 16 2018).

The line quoted ending in that signature is contributed by Colin Barker on Feb 26 2018.

The word *Empirical* — equivalently *Conjecture* — is the entry's own: the statement is recorded as fitted to the published terms, and no proof is given there. As of revision 8, last modified Feb 26 2018, the entry records no proof and links to none, so the statement is open. This note settles it.

Written out, the claim is

$$a(n) = 3\,a(n-1) - a(n-2) + 2\,a(n-3) + 4\,a(n-4) - a(n-5) + 2\,a(n-6) - \\ 2\,a(n-7) - 3\,a(n-8) - 2\,a(n-9) - 2\,a(n-10) - 2\,a(n-11).$$

stated for $n > 12$.

Written out, the claim is

$$\sum_{n\geq 1} a(n)\,x^n \;=\; \frac{4x + 13x^2 - x^3 + 12x^4 + 3x^5 - 13x^6 - 8x^7 - 19x^8 - 13x^9 - 7x^{10} - 2x^{11} - 4x^{12}}{1 - 3x + x^2 - 2x^3 - 4x^4 + x^5 - 2x^6 + 2x^7 + 3x^8 + 2x^9 + 2x^{10} + 2x^{11}}.$$

2 The object

The name is read as follows. The reading is not a guess: it is pinned against the entry's own published terms, and against an independent enumeration, before anything is proved (Section 7).

Let $q = 2$ and let A be an $n \times 3$ array with entries in $\{0, 1, \dots, 1\}$, rows indexed $0 \le i < n$ and columns $0 \le j < 3$.

The neighbour set the entry names by *king-move* is

$$\mathcal{N} \;=\; \{(-1,-1), (-1,0), (-1,1), (0,-1), (0,1), (1,-1), (1,0), (1,1)\}.$$

For a cell (i, j) of the array put

$$c(i,j) \;=\; \#\{(\delta,\varepsilon) \in \mathcal{N} \;:\; 0 \le i+\delta < n,\; 0 \le j+\varepsilon < 3,\; A[i+\delta][j+\varepsilon] = A[i][j]\},$$

the number of neighbours of the cell whose value is equal to the cell's own. Offsets that leave the array are not counted, so cells on the border have fewer neighbours than interior ones — which is what makes the two readings of the entry's wording distinguishable on the entry's own data.

The condition is $c(i,j) \in \{0, 1, 2, 3, 6\}$ for every cell of the array, together with $A[0][0] = 0$.

$a(n)$ is the number of such arrays. The entry's offset is 1, so its term of index n is the count for n rows.

3 The count is a walk count

Every offset in $\mathcal{N}$ moves at most one row, so whether a cell (i, j) satisfies its condition is decided by rows $i-1$, i and $i+1$ alone. A window of three consecutive rows therefore decides the whole of its middle row.

Let $R = \{0, \dots, 1\}^3$ be the set of rows, so $|R| = 8$, and adjoin a symbol $\top$ standing for "no row above". Take as states the pairs

$$\Sigma \;=\; (R \cup \{\top\}) \times R,$$

with an edge from (p, c) to (c, x) exactly when row c satisfies the condition in the window (p, c, x), and call (p, c) accepting when c satisfies the condition in the window $(p, c, \perp)$, where $\perp$ stands for "no row below".

An array with rows $r_1, \dots, r_n$ is then exactly a walk

$$(\top, r_1) \to (r_1, r_2) \to \cdots \to (r_{n-1}, r_n)$$

of length $n-1$ beginning at a state $(\top, r_1)$ with $r_1[0] = 0$ and ending at an accepting state. Every array gives one such walk and every such walk one array. With M the adjacency matrix, w the indicator of the allowed starting states and f that of the accepting ones,

$$a(n) \;=\; w^{\mathsf{T}} M^{n-1} f \qquad (n \ge 1),$$

so a satisfies the linear recurrence whose characteristic polynomial is that of M, and is in particular C-finite. The argument is the transfer-matrix method; see Stanley §4.7.

4 The degree bound, derived

The construction gives 72 states. Removing states unreachable from a start state, and states from which no accepting state can be reached, leaves 58; neither removal changes any count.

Two states from which the same number of arrays can be completed, for every remaining number of rows, contribute identically to $w^{\mathsf{T}} M^{n-1} f$ and may be identified. The coarsest such identification is computed by partition refinement: begin with the partition by acceptance and separate two states as soon as they send different numbers of edges into some block. The refinement terminates; on its blocks the number of completions of each length is constant by induction on that length, so the quotient digraph counts exactly what the original does.

That refinement leaves 14 blocks. The mirror reduction is then applied: two blocks receiving the same weight from the start, for every walk length, also contribute identically, and the refinement that finds them looks at edges coming in rather than going out. Writing L for the resulting $S \times S$ matrix with $S = 14$, $\tilde{w}$ for the start weights and $\tilde{f}$ for the acceptance weights,

$$a(n) \;=\; \tilde{w}^{\mathsf{T}} L^{n-1} \tilde{f} \qquad (n \geq 1),$$

and a satisfies the linear recurrence given by the characteristic polynomial of L, of degree 14. This is the only bound used below, and it is computed, not assumed.

5 The decision procedure

Let $c_1, \ldots, c_D$ be the coefficients of a claimed recurrence $a(n) = \sum_{i=1}^{D} c_i\, a(n-i)$, let $q(t) = t^D - \sum_{i=1}^{D} c_i t^{D-i}$ be its characteristic polynomial, and put

$$u_m \;=\; \tilde{w}^{\mathsf{T}} L^{m-1} q(L)\, \tilde{f} \qquad (m \geq 1).$$

Expanding the powers of L and using the displayed formula for a, one gets $u_m = a(n) - \sum_{i=1}^{D} c_i\, a(n-i)$ with $n = m+D+0$. So u_m is exactly the residual of the claim at $n = m+D+0$, and the recurrence holds at an index n if and only if the corresponding u_m vanishes.

Now $u_m = \tilde{w}^{\mathsf{T}} L^{m-1} y$ with $y = q(L)\tilde{f}$ a fixed vector, so (u_m) satisfies the characteristic polynomial of L, of degree 14: writing that polynomial as $t^{14} - \sum_{i=1}^{14} \lambda_i t^{14-i}$ gives $u_{m+14} = \sum_{i=1}^{14} \lambda_i u_{m+14-i}$. Hence 14 consecutive vanishing residuals force every later residual to vanish, by induction. The test is therefore finite; and because it locates the *last* nonvanishing residual, it pins the threshold exactly rather than merely bounding it.

Everything is carried out in exact integer arithmetic on the vector $L^{m-1} y$. The matrix M is never formed.

6 Results

Theorem 1. *For A298282, the recurrence* $a(n) = 3\,a(n-1) - a(n-2) + 2\,a(n-3) + 4\,a(n-4) - a(n-5) + 2\,a(n-6) - 2\,a(n-7) - 3\,a(n-8) - 2\,a(n-9) - 2\,a(n-10) - 2\,a(n-11)$ *holds* *for every $n > 12$, and fails at $n = 12$.*

Proof. The residuals u_m were computed exactly for $m = 1, \ldots, 56$. The last nonvanishing one is at $m = 1$, that is at $n = 12$. After it, more than $S = 14$ consecutive residuals vanish, so by Section 5 every later residual vanishes as well. □

The entry claims the recurrence for $n > 12$. The theorem is at least as strong.

Theorem 2. *For A298282, the generating function is exactly* $\sum_{n\geq 1} a(n)x^n = \frac{4x+13x^2-x^3+12x^4+3x^5-13x^6-8x^7-19x^8-}{1-3x+x^2-2x^3-4x^4+x^5-2x^6+2x^7+3x^8}$

Proof. Let $N(x)$ and $D(x)$ be the numerator and denominator above, $D(0) = 1$. The residual test applied to the recurrence read off D — namely $a(n) = \sum_i c_i a(n-i)$ with c_i the negated coefficients of D — gives vanishing residuals for every $n > 12$, so the power series $F(x) = \sum_{n\geq 1} a(n)x^n$ satisfies $[x^k](F(x)D(x)) = 0$ for every $k > 13$. The remaining coefficients of $F(x)D(x)$, for $k \leq 26$, were computed from the walk count and agree with $N(x)$ term by term. Hence $F(x)D(x) = N(x)$ identically, which is the assertion. □

7 Verification

Four checks were run, each reproducible from the code accompanying this note, and the first two are what pin the reading of the entry's English.

- **The published terms.** The walk count reproduces all 25 terms the entry publishes, exactly, in integer arithmetic, with the index taken from the entry's stated offset (1) rather than fitted. The first of them are 4, 25, 70, 205, 614, 1860, 5631, 17034, ...
- **An independent enumeration.** For $n = 1, \ldots, 5$ every one of the 2^{3n} arrays was written out and tested cell by cell against the condition as the entry states it — no transfer matrix, no row window, a separate piece of code. The counts agree with the walk count and with the entry.
- **The claim on the raw data.** Each claim was evaluated on the entry's published terms alone, with no model involved, wherever the proved range and the published data overlap. It holds there.
- **The annihilation test.** Carried to the derived bound $S = 14$ over the whole vector, in exact integer arithmetic, never sampled and never truncated.

8 References

1. OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A298282, revision 8, last modified Feb 26 2018.
2. R. P. Stanley, *Enumerative Combinatorics*, Vol. 1, 2nd ed., Cambridge University Press, 2012; §4.7, the transfer-matrix method.
3. M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011; C-finite sequences and their closure properties.

4. J. Berstel and C. Reutenauer, *Noncommutative Rational Series with Applications*, Cambridge University Press, 2011; linear representations of counting automata and their reduction.